

CMS Open Data H to tau tau Search in the mu tauh Final State: Phase 4a Expected Results

Analysis my_analysis

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Contents

Change Log	2
1 Introduction	2
2 Data Samples	2
3 Event Selection	5
4 Template Construction and Statistical Model	13
5 Systematic Uncertainties	14
5.0.1 Luminosity Normalization	14
5.0.2 DY/Z Normalization	15
5.0.3 Tau Trigger, Tau ID, and Open-Data Acceptance	15
5.0.4 MC Statistical Uncertainty	16
5.0.5 Downscoped Systematic Sources	16
6 Cross-Checks and Validation	17
7 Expected Results	20
8 Comparison to Prior Results and Theory	23
9 Conclusions	23
10 Future Directions	24
11 Known Limitations and Open Questions	24
12 Appendix A: Limitation Index	24
13 Appendix B: Numerical Source Map	25
14 Appendix C: Auxiliary Phase 2 Figures	25
15 Appendix D: Auxiliary Phase 3 Template Figures	30
16 Appendix E: Reproduction Contract	32
17 Appendix F: Note Writer Self-Check	33
References	34

Change Log

Phase 4a v1

- Initial AN version (Phase 4a).
- Added the expected-only visible-mass template-fit note for the reduced CMS 2012 Open Data H to tau tau search in the mu tauh final state.
- Added the Asimov expected CLs limit, discovery diagnostic, signal-injection validation, limited GoF toy study, systematic program, missing-component limitations, and reproduction contract.
- Recorded that 10% data validation and full observed signal-region results are intentionally absent until the Phase 4b and Phase 4c gates.

1 Introduction

This note documents a reduced-sample search for a Standard Model Higgs boson decaying to a pair of tau leptons in CMS 2012 Open Data. The analyzed final state contains one reconstructed muon and one hadronically decaying tau candidate, denoted mu tauh in this note. The analysis follows the logic of the CMS Run 1 H to tau tau searches while remaining explicit about the limitations of the reduced outreach NanoAOD-like files used here (CMS Collaboration 2014, 2018, 2021a).

The primary result in this Phase 4a document is expected-only. The fit observation is background-only Asimov pseudo-data derived from the nominal Monte Carlo templates, not the full collision-data signal-region distribution. The observed 10% validation result will be added in Phase 4b, and the full observed signal-region result will be added only after the Phase 4b human unblinding gate.

The Phase 1 strategy decision [D1] defines this work as a search/template-fit analysis rather than a measurement. The parameter of interest is the signal-strength modifier μ , which scales the Standard Model H to tau tau signal templates. Because the reduced sample set is single-channel, uses visible mass rather than the full CMS SVFit reconstruction, and lacks several paper-level background and calibration components, the expected sensitivity reported here is a reduced-open-data benchmark rather than an attempt to reproduce the all-channel CMS Run 1 result.

The current model fits the visible mu tauh mass, m_{vis} , simultaneously in three mutually exclusive categories: VBF, boosted, and zero-jet. The Phase 4a expected median 95% CLs upper limit is $\mu = 11.374$, with an Asimov observed-on-background limit of $\mu = 11.374$. The discovery diagnostic on signal-plus-background Asimov data gives $Z = 0.191$, so this configuration is an expected-limit demonstration, not an expected-discovery analysis.

2 Data Samples

The data and simulation inputs are the reduced CMS Open Data H to tau tau files localized in Phase 2 from the public reduced ROOT mirror associated with the CERN Open Data H to tau tau outreach analysis (CMS Collaboration 2021a). Phase 2 resolved the data stream to TauPlusX Run2012B and Run2012C, not the prompt-listed SingleMu files. Phase 3 then required the TauPlusX trigger `HLT_IsoMu17_eta2p1_LooseIsoPFTau20` in all analysis regions.

The normalization uses the integrated luminosity value `L_int = 11.467/fb = 11467/pb` recorded in `phase3_selection/outputs/normalization_inputs.json`. That JSON traces the value to the CMS Open Data H to tau tau tutorial `skim.cxx` and to the 2012 CMS open-data luminosity records (CMS Open Data Analyses 2021; CMS Collaboration 2017). The luminosity uncertainty is 2.6%, sourced from CMS PAS LUM-13-001 and implemented as a correlated normalization nuisance for MC-normalized predictions (CMS Collaboration 2013).

Table 1: Collision-data sample summary. The official event counts and luminosities come from `phase3_selection/outputs/normalization_inputs.json`, which records CERN Open Data records 12358 and 12359. Local entries are reduced processing entries in the localized ROOT files and are not used as luminosity inputs.

Period	Data file	Stream	Official events	Local entries	Luminosity [/fb]	Record
2012B	Run2012B_TauPlusX.root	TauPlusX	35,647,508	3,564,750	4.4117	12358 (CMS Collaboration 2021d)
2012C	Run2012C_TauPlusX.root	TauPlusX	51,303,171	5,130,317	7.0552	12359 (CMS Collaboration 2021e)
B+C	combined	TauPlusX	86,950,679	8,695,067	11.4670	tutorial context (CMS Collaboration 2021a; CMS Open Data Analyses 2021)

The simulation samples cover the two available H to tau tau signal production modes and the dominant available backgrounds. The MC denominator for every simulated process is the official CERN Open Data `distribution.number_events` value recorded in `normalization_inputs.json`; it is not the local reduced `Events` tree entry count, and it is not back-calculated from data. Signal normalization uses the production cross section times the H to tau tau branching fraction because the signal files are decay-filtered H to tau tau samples.

Table 2: Simulation sample summary and absolute weights. Cross sections, generated-event denominators, record identifiers, and per-local-entry weights are read from `phase3_selection/outputs/normalization_inputs.json`. The Higgs signal cross sections are H to tau tau effective cross sections and are traced in the JSON to the LHC Higgs Cross Section Working Group input (LHC Higgs Cross Section Working Group 2017).

Sample	Process	Generator	Cross section [pb]	N_gen	Weight	Record
GluGluToHToTauTau	ggH H to tau tau	Powheg+Pythia6	1.338	476,963	0.03217	12351 (CMS Collaboration 2021c)
VBH_HToTauTau	VBH H to tau tau	Powheg+Pythia6	0.1001	491,653	0.002335	12352 (CMS Collaboration 2021g)
DYJetsToLL	Z/gamma* to ll	MadGraph+Pythia6	3503.7	30,458,871	1.319	12353 (CMS Collaboration 2021b)
TTbar	ttbar	MadGraph+Pythia6	252.9	6,423,106	0.4515	12354 (CMS Collaboration 2021f)
W1JetsToLNu	W to lnu + 1 jet	MadGraph+Pythia6	6381.2	29,784,800	2.457	12355 (CMS Collaboration 2021h)
W2JetsToLNu	W to lnu + 2 jets	MadGraph+Pythia6	2039.8	30,693,853	0.7621	12356 (CMS Collaboration 2021i)
W3JetsToLNu	W to lnu + 3 jets	MadGraph+Pythia6	612.5	15,241,144	0.4608	12357 (CMS Collaboration 2021j)

The broader CMS H to tau tau analyses use additional ingredients that are not present in the reduced mirror. The missing or deferred components include embedded Z to tau tau, QCD multijet samples, diboson, single top, W4 and inclusive W+jets, associated Higgs production, H to WW, additional DY categories, and Run2012A/D TauPlusX. These are not silently substituted in this Phase 4a expected model; their impacts are carried explicitly in Table 3 and in the limitation index.

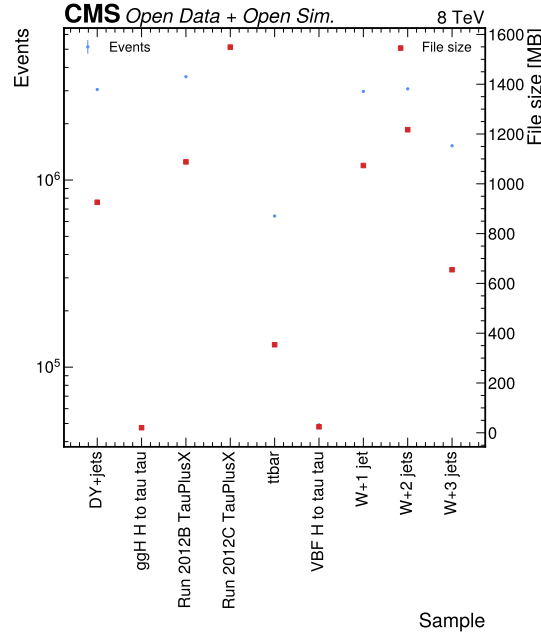


Figure 1: Sample size and local-entry inventory. The Phase 2 plot summarizes local reduced ROOT `Events` entries and file sizes for the localized H to tau tau inputs. The analysis uses these local entries for processing and template filling, while official Open Data record counts from Table 2 provide MC normalization denominators.

Table 3: Missing or deferred paper-level components. The entries are read from `phase4_inference/4a_expected/outputs/1imitations_downscope.json`, which propagates the Phase 2 final-AN sample-limitation obligations.

Component	Role	Status	Expected Phase 4a impact
Run2012A_TauPlusX	data	deferred-AOD-conversion	Not included in the reduced B+C luminosity and observed-data validation scope.
Run2012D_TauPlusX	data	deferred-AOD-conversion	Not included in the reduced B+C luminosity and observed-data validation scope.
DYJetsToTauTau__embedded	background	missing-reduced	DY tau response cannot use the embedded-sample method of the CMS reference analyses.
DYJets_ZToMuMu_EE__fakes	background	missing-reduced	Separate lepton-to-tau fake components are not modelled as independent templates.
QCD_Multijet	background	missing-reduced	Instrumental fake-tau background is deferred to sideband validation phases.
WW, WZ, ZZ	background	deferred-AOD-conversion	Diboson contributions are omitted from the expected workspace.
SingleTop	background	deferred-AOD-conversion	Single-top contribution is omitted from the expected workspace.
W4JetsToLNu,	background	missing-reduced	W+jets stitching is limited to W1/W2/W3 samples.
WJetsToLNu_inclusive	background	missing-reduced	DY modelling uses the inclusive reduced
DY_jet/mass-binned categories	background	missing-reduced	DYJetsToLL file.
WH, ZH, ttH H to tau tau	signal	deferred-AOD-conversion	Associated-production signal acceptance is not part of the signal model.
HToWW	background	deferred-AOD-conversion	H to WW contamination is not modelled as a separate component.

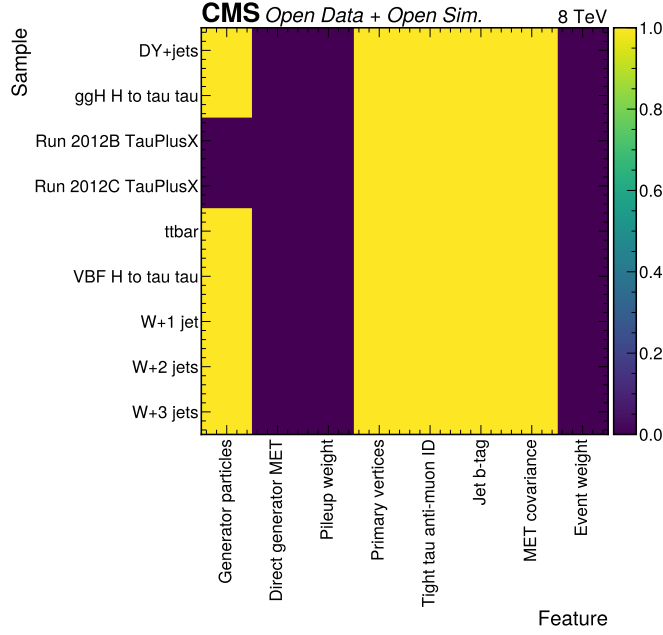


Figure 2: Feature availability in the reduced files. The Phase 2 heatmap records whether the data and MC files contain the branch classes needed for trigger, object selection, jets, MET, and generator studies. It motivates the current reduced-sample scope: the mu tauh selection and VBF categories are feasible, while direct GenMET, event weights, and pileup weights are absent.

3 Event Selection

The event selection follows the Phase 3 output and is summarized here because the Phase 4a templates are built directly from the Phase 3 selected-event summary. All signal, control, and validation regions require `HLT_IsoMu17_eta2p1_LooseIsoPFTau20`. The available single-muon triggers `HLT_IsoMu24` and `HLT_IsoMu24_eta2p1` are not ORed into the analysis because they define a different higher- p_T muon phase space.

Muon candidates require transverse momentum greater than 20 GeV, pseudorapidity magnitude below 2.1, tight muon ID, and finite relative isolation below 0.20. Tauh candidates require transverse momentum greater than 20 GeV, pseudorapidity magnitude below 2.3, decay-mode ID, tight isolation ID, the tight anti-muon veto, finite tau relative isolation, and separation from the selected muon. Jets require transverse momentum greater than 30 GeV, pseudorapidity magnitude below 4.7, and separation from the selected leptons.

The signal-region candidate set requires an opposite-sign muon-tauh pair and transverse mass $m_T(\mu, MET) < 40 \text{ GeV}$. The W control handle uses opposite-sign events with $m_T(\mu, MET) > 80 \text{ GeV}$; the QCD sideband uses same-sign low- m_T events; the Z-rich validation handle uses opposite-sign low- m_T events with $60 < m_{\text{vis}} < 120 \text{ GeV}$; and the top handle uses events with a positive valid b-tag score. The Phase 4a expected model uses only the signal-region category templates, while the control and validation handles remain documented inputs for Phase 4b and Phase 4c.

Signal-region events are assigned to exactly one fit category. The VBF category is assigned first using at least two clean jets, leading-dijet mass above 300 GeV, and leading-dijet absolute pseudorapidity separation above 2.5. Non-VBF events with at least one clean jet enter the boosted category, and the remaining non-VBF events enter the zero-jet category.

Table 4: Phase 3 raw selected counts by category and sample. Counts are read from `phase3_selection/outputs/category_yields.json` and are unweighted processing counts, not luminosity-normalized yields.

Sample	Role	VBF	Boosted	Zero-jet
Run2012B_TauPlusX	data	34	858	3,183
Run2012C_TauPlusX	data	45	1,355	5,283
GluGluToHToTauTau	signal	23	249	439
VBF_HToTauTau	signal	364	499	94
DYJetsToLL	background	34	802	4,432
TTbar	background	170	1,513	88
W1JetsToLNu	background	4	37	350
W2JetsToLNu	background	16	238	348
W3JetsToLNu	background	16	228	126

Table 5: Phase 3 control and validation raw counts. The data totals are from Run2012B and Run2012C combined, and the signal and background totals are summed over the available reduced MC files. These regions are not used to unblind signal-region results in Phase 4a.

Region or handle	Data	Signal MC	Background MC	Phase 4a role
Signal-region candidate	10,758	1,668	8,402	Template source for expected model
W high-mT	4,023	62	6,283	Deferred W normalization/validation handle
QCD same-sign	3,138	34	1,200	Deferred instrumental-background sideband
Z-rich	5,514	1,261	4,396	DY/Z validation handle
Top b-tag handle	6,689	1,150	12,508	Top-enriched validation handle

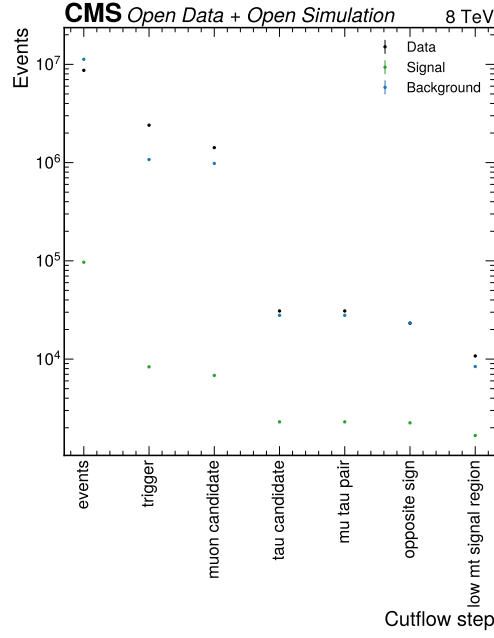


Figure 3: Phase 3 cutflow summary. The plot shows raw event counts after each selection step, grouped by data, signal MC, and background MC. The cutflow is monotonic and demonstrates that the reduced TauPlusX trigger, object IDs, tight tau anti-muon veto, opposite-sign requirement, and low-mT selection leave populated inputs for the expected fit.

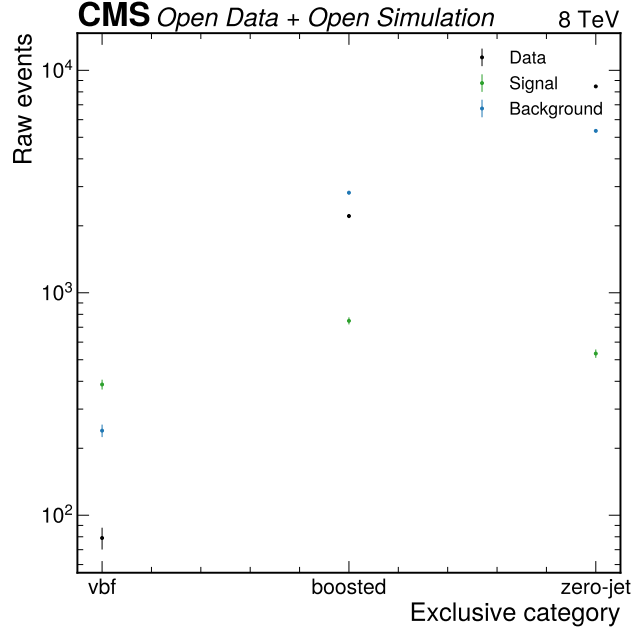


Figure 4: Exclusive category yields. The figure shows raw selected event counts in the VBF, boosted, and zero-jet categories retained for the simultaneous fit. It verifies that the VBF category remains populated after the trigger fix, while the boosted and zero-jet categories provide the dominant event counts.

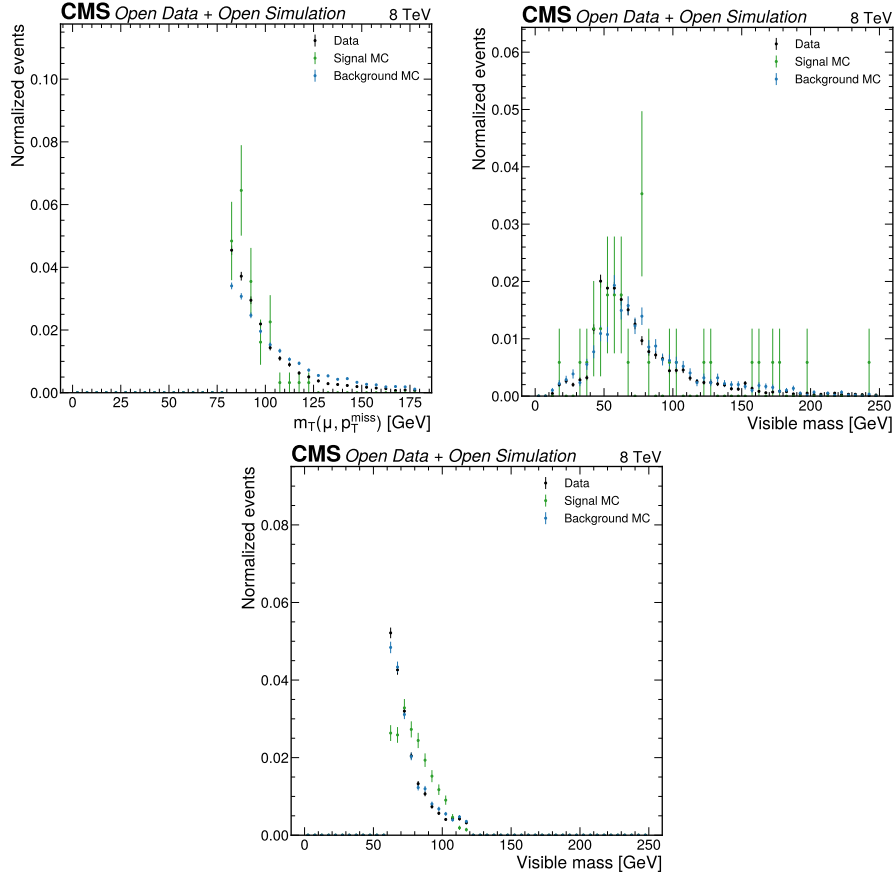


Figure 5: Phase 3 control and validation diagnostics. From left to right, the panels show the W high- m_T control-region shape comparison, the same-sign QCD sideband visible-mass comparison, and the Z-rich validation visible-mass comparison. These diagnostics document the available W+jets, fake-tau, and DY/Z handles while the Phase 4a expected model retains the corresponding limitations and does not tune these components to real data.

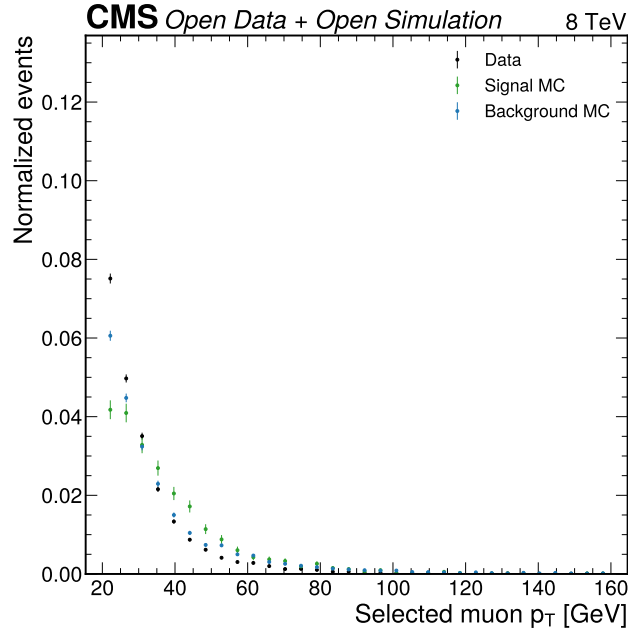


Figure 6: Selected muon transverse momentum. The Phase 3 plot uses final selected muons after trigger, ID, isolation, object arbitration, and signal-region requirements. It documents the muon kinematic phase space entering the Phase 4a templates rather than the looser first-candidate survey from Phase 2.

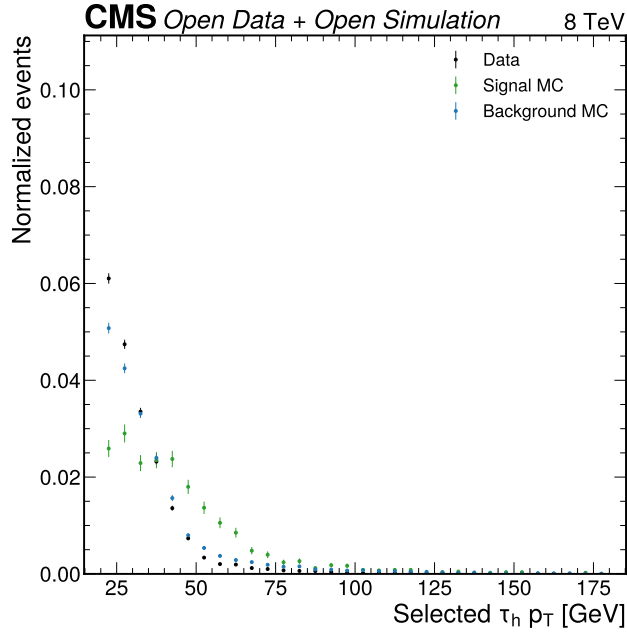


Figure 7: Selected tauh transverse momentum. The Phase 3 plot verifies that the tight tau isolation and tight anti-muon veto leave a physically populated tauh spectrum. This object definition is inherited without modification by the Phase 4a expected model.

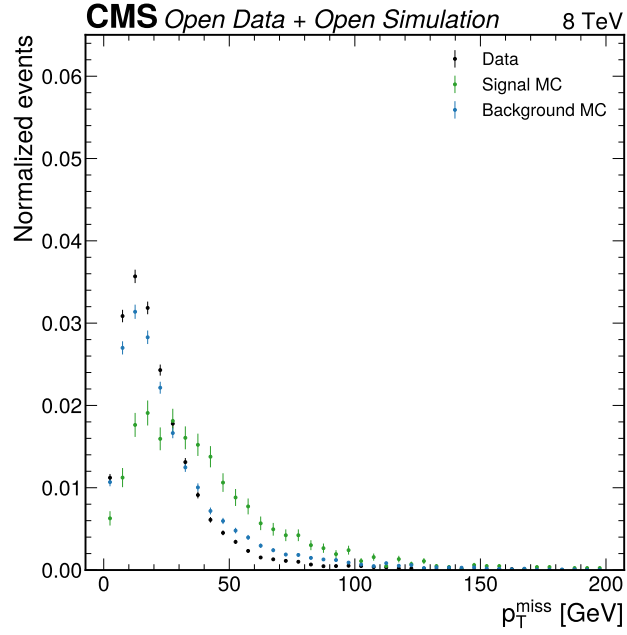


Figure 8: Missing transverse momentum. The Phase 3 MET comparison documents the reconstructed missing-momentum input used for transverse mass and add-MET diagnostics. It also illustrates why MET-sensitive alternative observables remain cross-checks rather than promoted primary fit variables in this reduced model.

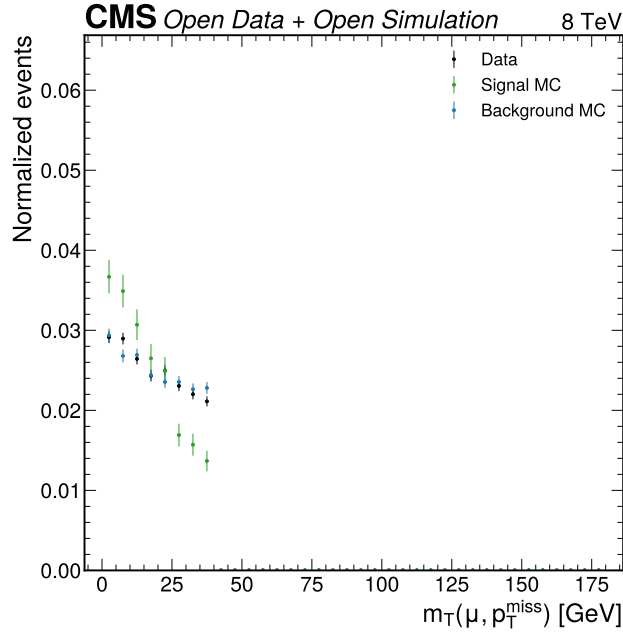


Figure 9: Muon-MET transverse mass. This distribution is shown after the low- m_T signal-region requirement and documents the variable that separates the signal-region candidate set from the high- m_T W control handle. The same quantity is deliberately not used as an MVA discriminant after Phase 3 validation failures in the broader input set.

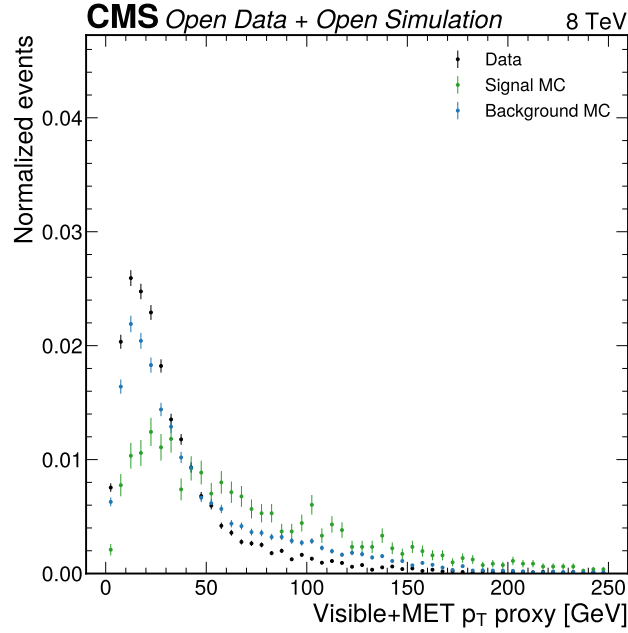


Figure 10: Visible-plus-MET transverse momentum proxy. This Phase 3 diagnostic records the available proxy for the boosted topology. The raw data/MC modelling was not sufficient for MVA promotion, but the variable remains part of the category and alternative-observable context.

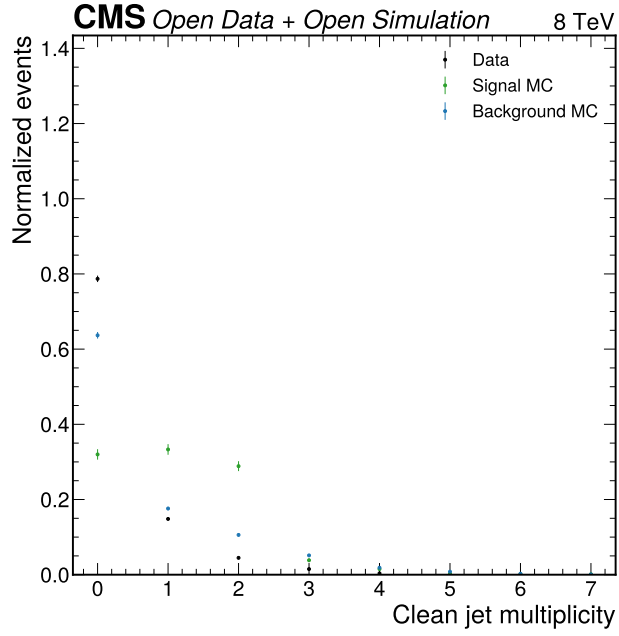


Figure 11: Clean jet multiplicity. The figure motivates the three retained fit categories by showing the selected-event jet multiplicity. The visible data/MC shape difference is one reason the analysis uses transparent category boundaries and avoids a trained classifier in Phase 4a.

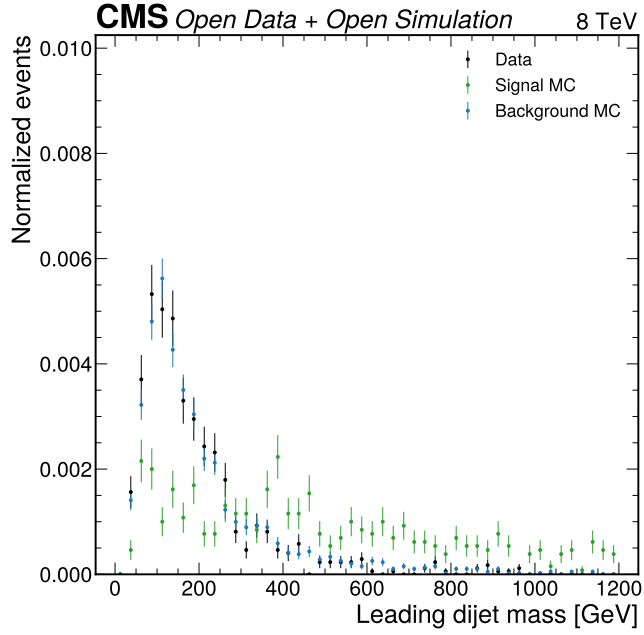


Figure 12: Leading dijet mass. The selected dijet-mass distribution supports the VBF-enriched category definition and was one of the few candidate MVA inputs to pass the Phase 3 modelling gate. The category threshold at 300 GeV is therefore retained as a transparent VBF-like requirement.

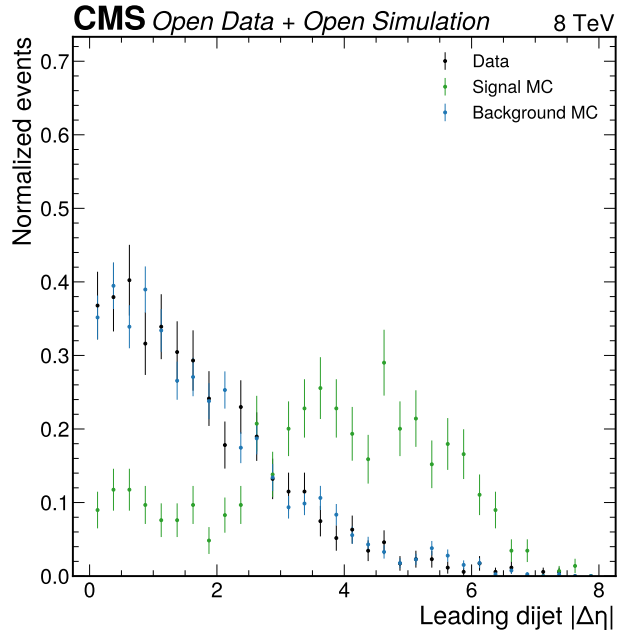


Figure 13: Leading dijet pseudorapidity separation. This selected-event distribution supports the VBF category requirement on the leading-dijet absolute eta separation. It also passed the Phase 3 MVA input-quality gate, but the classifier path was still downscoped because most other inputs failed.

4 Template Construction and Statistical Model

The Phase 4a statistical model is a binned template fit in m_{vis} . The bin edges are 0, 40, 60, 80, 100, 120, 160, and 250 GeV, read from `phase4_inference/4a_expected/outputs/nominal_yields.json`. The categories are VBF, boosted, and zero-jet, and the observation vector is the background-only Asimov expectation.

The per-entry MC weight for a background sample s is

$$w_s = \frac{\sigma_s L_{int}}{N_{gen,s}}.$$

For signal samples, the stored cross section is the H to tau tau effective cross section, so the same expression is written in the source JSON as

$$w_s = \frac{\sigma_{prod,s} B(H \rightarrow \tau\tau) L_{int}}{N_{gen,s}}.$$

The expected bin yield in category c and visible-mass bin b is the sum of weighted entries over the samples that belong to a process group:

$$\nu_{cb}(\mu, \theta) = \mu \sum_{s \in S} \nu_{cb}^{sig}(\theta) + \sum_{p \in B} \nu_{cb}^{bkg}(\theta).$$

Here μ is the signal-strength modifier, θ denotes nuisance parameters, S denotes the two available Higgs signal samples, and B denotes the available background processes. The nominal background-only Asimov observation is

$$n_{cb}^A = \nu_{cb}(\mu = 0, \theta = 0).$$

The likelihood is a HistFactory-style product of Poisson terms and nuisance constraints implemented with pyhf version 0.7.6 (Heinrich et al. 2021). In schematic form,

$$L(\mu, \theta) = \prod_{c,b} \text{Pois}(n_{cb} | \nu_{cb}(\mu, \theta)) \prod_k \pi_k(\theta_k).$$

The profile likelihood ratio for a tested signal strength is

$$\lambda(\mu) = \frac{L(\mu, \hat{\hat{\theta}}_\mu)}{L(\hat{\mu}, \hat{\theta})},$$

where $\hat{\hat{\theta}}_\mu$ are the profiled nuisance parameters at fixed μ , and hats without double hats denote the unconditional maximum-likelihood estimates. Upper limits use the modified frequentist CLs construction (Read 2002) with asymptotic formulae (Cowan et al. 2011),

$$CL_s(\mu) = \frac{p_{s+b}(\mu)}{1 - p_b(\mu)}.$$

The reported 95% upper limit is the value of μ for which

$$CL_s(\mu) = 0.05.$$

The discovery diagnostic uses the one-sided q_0 statistic, with the convention that downward fitted signal strengths do not count as discovery evidence:

$$q_0 = \begin{cases} -2 \ln \lambda(0), & \hat{\mu} \geq 0, \\ 0, & \hat{\mu} < 0. \end{cases}$$

The signal-injection validation creates Asimov pseudo-data at specified injected signal strengths and refits μ . The bias is computed as

$$\Delta\mu = \hat{\mu}_{fit} - \mu_{inj},$$

and the relative bias for nonzero injections is

$$\delta_\mu = \frac{\hat{\mu}_{fit} - \mu_{inj}}{\mu_{inj}}.$$

The GoF plumbing check uses a saturated statistic for Poisson templates. For an observed or pseudo-observed count n_b and expectation ν_b , the per-bin contribution is

$$t_b = 2 \left[\nu_b - n_b + n_b \ln \left(\frac{n_b}{\nu_b} \right) \right],$$

with the logarithmic term set to zero when $n_b = 0$. The Phase 4a Asimov saturated statistic is identically zero by construction because $n_b^A = \nu_b$, so the GoF toy distribution validates toy generation and statistic plumbing rather than data/model agreement.

Table 6: Phase 4a workspace configuration. Values are read from `expected_results.json`, `nominal_yields.json`, and the Phase 4a workspace metadata.

Quantity	Value	Source
pyhf version	0.7.6	<code>expected_results.json</code>
Parameter of interest	mu	<code>expected_results.json</code>
Number of parameters	25	<code>expected_results.json</code>
Fit categories	3	<code>nominal_yields.json</code>
Visible-mass bins per category	7	<code>nominal_yields.json</code>
POI scan range	0 to 50	<code>expected_results.json</code>
Observation source	background-only Asimov	<code>expected_results.json</code>

5 Systematic Uncertainties

The Phase 4a systematic program is deliberately narrower than the CMS reference analyses. It implements only sources that can be justified from the reduced open-data inputs and the predeclared Phase 1 strategy. Detector-shape variations requiring unavailable scale-factor machinery or shifted reduced inputs are documented as downscoped limitations instead of being replaced by arbitrary variations.

5.0.1 Luminosity Normalization

The luminosity nuisance represents the normalization uncertainty on all MC-based predictions. Its physical origin is the uncertainty in the integrated luminosity assigned to the 2012 Run2012B+C TauPlusX data sample. The implemented size is 2.6%, read from `systematics.json` and traced through `normalization_inputs.json` to CMS PAS LUM-13-001 (CMS Collaboration 2013).

The evaluation is a correlated normalization variation. For each affected template, the nuisance scales the nominal expected yield by a multiplicative factor corresponding to the luminosity uncertainty. In the schematic yield expression, the variation propagates through the luminosity factor in the MC weight:

Table 7: Implemented Phase 4a nuisance parameters. Values are read from `phase4_inference/4a_expected/outputs/systematics.json`.

Nuisance	Type	Size	Applies to	Source in artifact
lumi_2012	normsys	2.6%	all MC-normalized samples	CMS PAS LUM-13-001
dy_norm_open_data	normsys	15%	DYJetsToLL	Phase 1 [D6] and user 10-15% requirement
tau_open_data_acceptance	normsys	15%	all selected MC templates	Phase 1 [D5]/[L2] and CMS Run 1 tau range
per-category statererror	statererror	sqrt(sum weights squared) per bin	all selected MC templates	finite selected MC counts and official weights

Table 8: Systematic completeness against search conventions and CMS references. The row content is copied from `systematics.json` and maps the current reduced-sample implementation to `conventions/search.md`, CMS JHEP 05 (2014) 104, and CMS Phys. Lett. B 779 (2018) 283.

Source	Convention	CMS 2014	CMS 2018	This analysis	Status
Signal cross-section theory	required	scale/PDF/UE/PS	signal theory and acceptance	normalization source recorded; dedicated theory NP downscopied	downscopied
Signal acceptance/shape	required	tau/JES/MET/generator shape	shape uncertainties in categories	tau/open-data acceptance rate NP only	partial
Luminosity	pp normalization	2.6% at 8 TeV	luminosity NP	2.6% normsys on MC samples	implemented
DY/Z normalization	background norm	inclusive Z plus category extrapolation	DY control constraints	15% normsys on DYJetsToLL	implemented
Tau ID/trigger/open-data acceptance	object calibration	tau ID/trigger range	tau ID/trigger effects	15% correlated rate NP	implemented
MC statistics	required	limited event counts	bin-by-bin MC stat	per-category statererror modifiers	implemented
QCD multijet	background norm/shape	data-driven estimate	data-driven estimate	missing reduced sample; sideband deferred	downscopied
W+jets transfer/control	background norm	high-mT control region	mT control region	W1/W2/W3 MC only in expected SR model	downscopied
Diboson and single top	pp electroweak backgrounds	included	included	no reduced samples; omitted	downscopied

$$w_s(\theta_L) = w_s(0)(1 + 0.026\theta_L).$$

The numerical impact is visible in the expected nuisance summary figure and in the workspace configuration: `lumi_2012` is one of the rate constraints in the 25-parameter pyhf model. Its impact is subdominant relative to the larger open-data acceptance and DY normalization nuisances because the luminosity uncertainty is much smaller. It remains mandatory because it is a real external normalization uncertainty on simulation-normalized predictions.

5.0.2 DY/Z Normalization

The DY/Z normalization nuisance covers the dominant Z/gamma* background in the visible-mass fit. Its physical origin is the incomplete reduced-open-data calibration path: the available reduced files lack the full trigger turn-on, tau efficiency scale-factor, and embedded Z to tau tau machinery used in the CMS reference analysis. Phase 1 decision [D6] required a 10–15% DY/Z normalization uncertainty before unblinding, and Phase 4a implements the high end, 15%, as `dy_norm_open_data`.

The variation applies only to the DYJetsToLL template. It is not derived by scaling DY to the observed signal region or by fitting the Drell-Yan peak in full data. In the model, the DY contribution in each category and bin is multiplied by a constrained factor:

$$\nu_{cb}^{DY}(\theta_{DY}) = \nu_{cb}^{DY}(0)(1 + 0.15\theta_{DY}).$$

The numerical impact is largest in the zero-jet category, where DY contributes most of the background total. This source is physically expected to matter because the zero-jet background total is 7060.345 expected events and the DY-rich validation handle contains 5514 data events at Phase 3. The interpretation is conservative but predeclared: Phase 4b must verify whether the 15% prior covers the 10% validation sample without making all pulls trivially small.

5.0.3 Tau Trigger, Tau ID, and Open-Data Acceptance

The tau/open-data acceptance nuisance represents missing efficiency and acceptance corrections for the reduced TauPlusX analysis. Its physical origin is broader than a single detector effect: the reduced files do not provide the full official tau trigger, tau ID, tau energy, and muon-tau trigger scale-factor machinery used in the CMS Run 1 analyses. Phase 1 [D5] and [L2] predeclared a 10–15% uncertainty range; Phase 4a implements 15% as `tau_open_data_acceptance`.

The evaluation is a correlated rate uncertainty on all selected MC templates. It is intentionally not a shape variation, because no citable reduced-sample detector-variation machinery or shifted template production exists at this phase. The propagated rate model is

$$\nu_{cb}^{MC}(\theta_\tau) = \nu_{cb}^{MC}(0)(1 + 0.15\theta_\tau).$$

The numerical impact is larger than luminosity and comparable in size to the DY normalization prior, but broader in scope because it affects signal and background MC. This is an honest reduced-sample limitation: it prevents overclaiming precision, but it cannot replace a true tau efficiency, trigger, and tau energy scale program. Phase 4b and Phase 5 must either source a more specific correction path or keep the limitation prominent.

5.0.4 MC Statistical Uncertainty

The MC statistical nuisance accounts for finite selected MC counts after category and visible-mass binning. Its physical origin is the limited number of reduced events in each process, especially for the small VBF category and for high-mass tail bins. The implementation uses per-category Barlow-Beeston-lite **staterror** modifiers in pyhf, with bin uncertainties derived from the weighted sum of squared MC weights.

For a bin filled by events with weights w_i , the MC statistical variance is

$$\sigma_{MC,b}^2 = \sum_i w_i^2.$$

The numerical impact is embedded in each category’s **staterror** modifier and summarized in **expected_nuisance_summary.pdf**. The smallest expected background bin remains above 10.141 events in the VBF category, so the asymptotic expected limit is not forced into a low-count toy-only regime by the final binning. The interpretation is that MC statistics are an implemented source, but the missing sample set can still bias the composition even when the finite-sample uncertainty on the included samples is represented.

5.0.5 Downscoped Systematic Sources

Several convention and reference-analysis sources are not implemented in Phase 4a because the reduced inputs do not support them without inventing variations. The downscoped list is read from **systematics.json**. These sources remain analysis limitations, not unmodelled surprises.

Table 9: Downscoped systematic and model sources. Each source is carried to the limitations section and Phase 4b/5 obligations.

Source	Reason	Impact
tau_energy_scale	no reduced tau energy scale variation or scale-factor prescription	could shift m_{vis} templates
muon_efficiency_scale	no official reduced-file muon trigger/ID scale factors	rate uncertainty partly covered but not separately measured
JES_JER_MET	no shifted reduced inputs or variation machinery	VBF and boosted acceptances may be understated
pileup	PV_npvs exists but no pileup weights/profile weights	qualitative validation limitation
PDF_scale_UE_PS	no event-level PDF/scale weights or alternative generators	theory and acceptance uncertainties are not separated
QCD_multijet	no reduced QCD MC and no Phase 4a data sideband tuning	instrumental fake component omitted from expected workspace
missing paper-level samples	no reduced files for several CMS components	background composition and sensitivity are incomplete

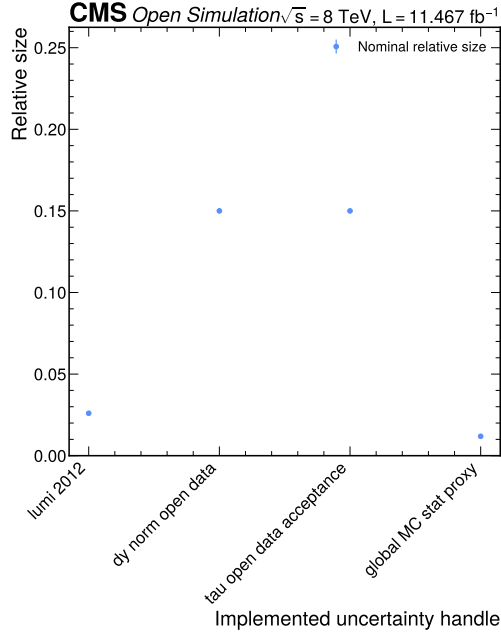


Figure 14: Expected nuisance summary. This Phase 4a diagnostic ranks implemented rate and MC-stat uncertainty handles in the expected workspace. It is not an observed-data post-fit impact plot; it shows the nuisance structure available before 10% validation and full unblinding.

6 Cross-Checks and Validation

The Phase 4a validation suite checks the expected model under its own assumptions. It does not claim data closure in the signal region, because the Phase 4a observation is Asimov pseudo-data. The checks therefore answer a narrower question: whether the expected workspace, scan, signal injection, and toy machinery behave coherently before data validation.

The Phase 3 validation remediation remains relevant context. The broad mixed validation/control union failed as a final background-only closure test, but separating the validation handles gave a Z-rich visible-mass χ^2 per degree of freedom of 1.409 after 4 remediation attempts. That result supports the current reduced DY shape handle while leaving W, QCD, top, and missing-background treatment as Phase 4b/4c validation obligations.

Table 10: Signal-injection recovery. Values are read from `phase4_inference/4a_expected/outputs/signal_injection.json`. The pass criterion is relative bias below 20% for nonzero injections and near-zero absolute bias for the zero injection.

Injected mu	Fitted mu	Abs. bias	Rel. bias	Pass
0.0	0.0000	0.0000	0.0000	true
1.0	1.0000	0.0000	0.0000	true
2.0	1.9989	-0.0011	-0.0006	true
5.0	4.9992	-0.0008	-0.0002	true

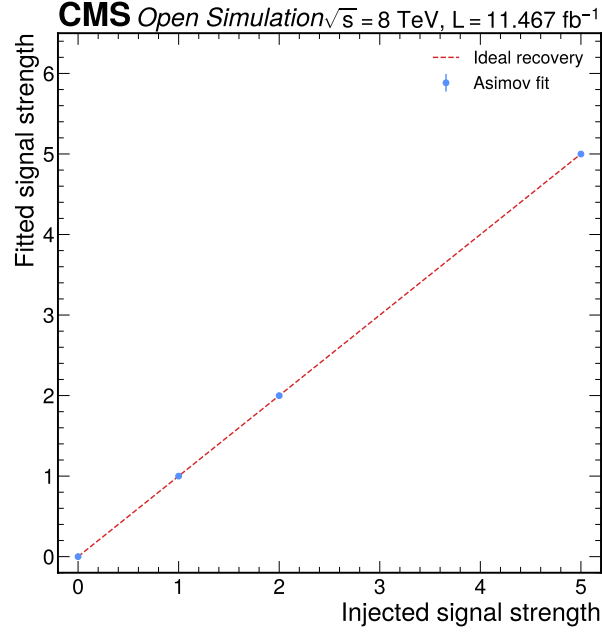


Figure 15: Signal-injection recovery. The Phase 4a injection study fits Asimov pseudo-data generated at injected signal strengths 0, 1, 2, and 5. The recovered signal strengths lie on the diagonal within the predeclared tolerance, which validates the workspace response under nominal model assumptions.

The GoF toy study uses 500 Poisson toys generated from the background-only expected templates. Since the Asimov observation equals the model expectation, the saturated statistic is 0.000 in every category and the toy fraction greater than or equal to the Asimov statistic is 1.000. This is a useful plumbing check but a deliberately limited validation: it cannot demonstrate agreement between real data and the model.

Table 11: Limited GoF toy summary. Values are read from `phase4_inference/4a_expected/outputs/gof_validation.json`. The p-like fraction is the fraction of Poisson toys with saturated statistic greater than or equal to the Asimov statistic.

Category	NDF	Asimov stat	Pearson chi2/ndf	Toy count	Toy mean	Toy median	Toy p95	Fraction $\geq$ Asimov
VBF	7	0.000	0.000	500	6.802	6.418	13.241	1.000
Boosted	7	0.000	0.000	500	6.933	6.579	13.096	1.000
Zero-jet	7	0.000	0.000	500	7.120	6.669	13.676	1.000

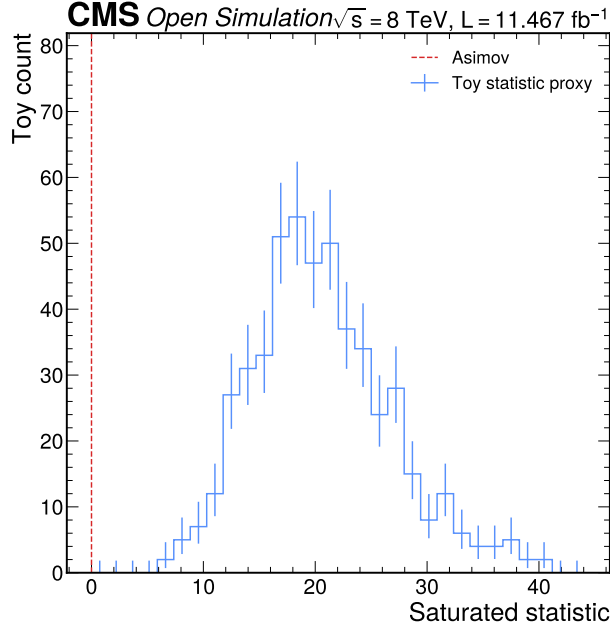


Figure 16: Limited toy GoF distribution. The toy distribution is generated from background-only expected templates and compared with the Asimov saturated statistic. The vertical marker at zero is expected by construction, so the figure validates toy generation and statistic evaluation rather than real-data agreement.

The alternative-observable gates were also evaluated before Phase 4a. Visible mass remained the primary observable because the Phase 3 raw MC-only separation metric was 60.63 for visible mass and 58.37 for add-MET mass, while the formal expected-sensitivity gate requires a nuisance-constrained statistical model. The NN classifier was downscoped because 13 of 16 candidate inputs failed the data/background-MC modelling gate, leaving only tau relative isolation, dijet mass, and dijet eta separation as passing inputs. The NN genMET regression was downscoped because the reduced MC files contain no direct GenMET branch and Phase 2 found no neutrino truth target.

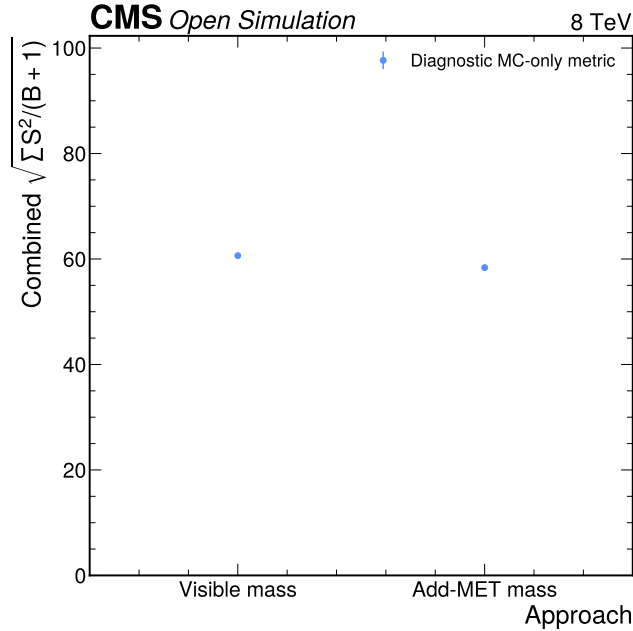


Figure 17: Approach comparison. The Phase 3 plot compares visible mass and add-MET mass using the same raw MC-only binned separation metric across the exclusive categories. Visible mass remains the Phase 4a primary because it has the better diagnostic metric and because add-MET has not passed a full nuisance-constrained expected-sensitivity replacement gate.

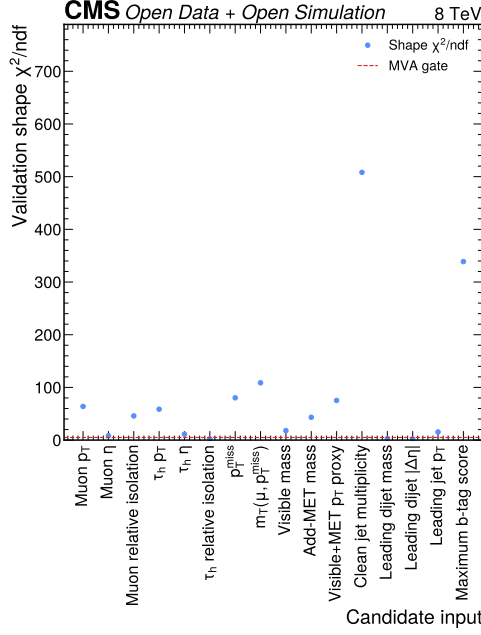


Figure 18: MVA input modelling gate. The Phase 3 gate compares validation/control-region data and background-MC shapes for candidate classifier inputs. Most candidate inputs exceed the predeclared χ^2/ndf threshold of 5, so the classifier is not trained or promoted in this Phase 4a model.

7 Expected Results

The expected-result tables and plots in this section are derived from `expected_results.json` and `nominal_yields.json`. They are Asimov/MC expected results only. No observed full-data signal-region result, observed signal strength, or observed full-data post-fit template is presented in this Phase 4a note.

The expected yields are small for signal relative to background in all categories. The VBF category has the best VBF enrichment but still contains only 1.590 expected signal events and 144.926 expected background events. The zero-jet category has the largest expected signal yield, 14.341, but it is embedded in 7060.345 expected background events.

Table 12: Nominal expected yields by category. Values are read from `phase4_inference/4a_expected/outputs/nominal_yields.json`.

Category	Signal yield	Background yield	S/sqrt(B)	Minimum background bin
VBF	1.590	144.926	0.132	10.141
Boosted	9.170	2093.690	0.200	101.333
Zero-jet	14.341	7,060.345	0.171	74.130

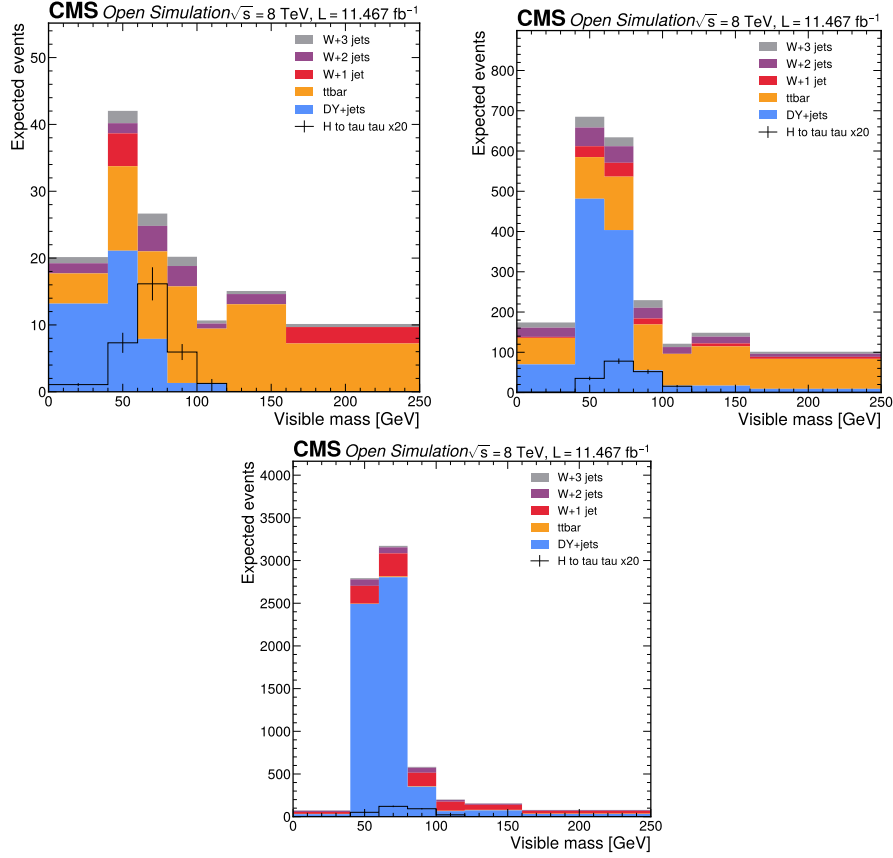


Figure 19: Expected visible-mass templates by fit category. From left to right, the panels show the VBF, boosted, and zero-jet categories with background-only expected stacks and nominal Higgs signal overlays after official Open Data normalization. The fit observation is the Asimov background expectation, not real collision data.

The asymptotic modified-frequentist CLs calculation scans μ from 0 to 50. The expected median 95% upper limit is 11.374, with the minus-two, minus-one, plus-one, and plus-two expected bands listed in Table 13. The Asimov observed-on-background limit is numerically equal to the median expected limit within rounding, as expected for a background-only Asimov observation.

The expected discovery diagnostic is intentionally weak. The pyhf asymptotic q_0 test on signal-plus-background Asimov data gives $p = 0.4244$ and $Z = 0.1906$. This is numerically compatible with a low-sensitivity expected-limit result rather than with an expected discovery claim, and it is consistent with the small expected $S/\sqrt{B}$ values in Table 12.

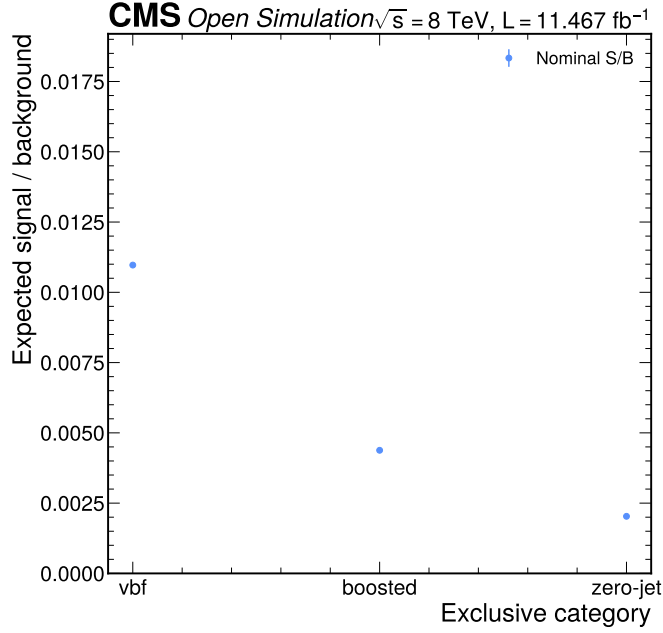


Figure 20: Expected signal-to-background ratio. This figure summarizes the category-integrated nominal Higgs signal divided by nominal background. The boosted category has the largest $S/\sqrt{B}$, while the VBF category remains important for topology coverage despite the small absolute signal yield.

Table 13: Expected 95% CLs upper limit on signal strength. Values are read from `phase4_inference/4a_expected/outputs/expected_results.json`.

Quantity	mu
Expected minus-two band	5.828
Expected minus-one band	7.958
Expected median	11.374
Expected plus-one band	16.472
Expected plus-two band	23.194
Observed-on-background Asimov	11.374

Table 14: Expected discovery diagnostic. Values are read from `expected_results.json`.

Method	p-value	Z	Status
pyhf asymptotic q0 on signal-plus-background Asimov	0.4244	0.1906	evaluated

8 Comparison to Prior Results and Theory

The most relevant Run 1 reference is CMS JHEP 05 (2014) 104, which reported evidence for H to tau tau using 7 TeV and 8 TeV data, multiple channels, CMS calibrations, embedded backgrounds, and a full category/control-region program (CMS Collaboration 2014). That publication quotes a global best-fit signal strength of 0.78 ± 0.27 and local significance above 3 standard deviations in the Higgs-mass range used in the strategy artifact. This Phase 4a reduced mu tauh expected limit of $\mu = 11.374$ is therefore weaker by a factor of about 14.6 relative to a Standard Model signal-strength target of 0.78, but the comparison is a sensitivity-scale context, not a direct observed-data pull.

CMS Phys. Lett. B 779 (2018) 283 reported H to tau tau observation at 13 TeV with $35.9/\text{fb}$, observed significance 4.9, expected significance 4.7, and signal strength 1.09 with uncertainties near 0.27 (CMS Collaboration 2018). The current reduced expected discovery diagnostic of $Z = 0.191$ is lower by a factor of about 24.6 relative to the 4.7 expected significance from that 13 TeV analysis. The comparison is quantitative but not a compatibility test because the energy, luminosity, channels, reconstruction, and training/calibration programs differ.

The ATLAS+CMS Run 1 combination reported a combined Higgs signal yield relative to the Standard Model of 1.09 ± 0.11 and an H to tau tau observed significance of 5.5 in the Phase 1 strategy’s reference table (ATLAS Collaboration and CMS Collaboration 2016). The Phase 4a expected limit is therefore not competitive with world-level Higgs coupling precision. Its purpose is to provide a transparent open-data search model with honest limitations and a reproducible statistical workflow.

Table 15: Prior-result context. External reference numbers are from the cited publications as recorded in the Phase 1 strategy, while this-analysis numbers are from Phase 4a JSON.

Source	Quantity	Value	Comparison status
This Phase 4a note	expected 95% CLs limit on mu	11.374	internal expected result
This Phase 4a note	expected discovery Z	0.191	internal expected diagnostic
CMS 2014 Run 1	global best-fit mu	0.78 ± 0.27	global observed context, not direct pull
CMS 2014 Run 1	local H to tau tau significance	> 3 sigma	global observed context
CMS 2018 13 TeV	observed significance	4.9	different energy and dataset
CMS 2018 13 TeV	expected significance	4.7	about 24.6 times larger than this expected Z
CMS 2018 13 TeV	signal strength	1.09	global observed context
ATLAS+CMS Run 1	combined signal yield relative to SM	1.09 ± 0.11	world-level coupling context
ATLAS+CMS Run 1	H to tau tau observed significance	5.5	global observed context

9 Conclusions

Phase 4a has produced an expected-only pyhf template model for the reduced CMS 2012 Open Data H to tau tau search in the mu tauh final state. The model uses visible mass in VBF, boosted, and zero-jet categories, official Open Data normalization metadata, a 2.6% luminosity nuisance, 15% DY and tau/open-data acceptance nuisances, and per-category MC statistical uncertainties. The expected median 95% CLs upper limit is $\mu = 11.374$, and the expected discovery diagnostic is $Z = 0.191$.

The result should be interpreted as a reduced-open-data expected sensitivity benchmark. It is not a full CMS reproduction, and it is not yet an observed result. Phase 4b will add 10% data validation with a compiled PDF and human gate, and Phase 4c will add full observed signal-region results only after approval.

10 Future Directions

The next required step is Phase 4b 10% data validation. That phase must compare the 10% observed result to the expected model, validate the Z-rich, W high-mT, QCD same-sign, top, VBF, and category-stability handles, and recompile the analysis note PDF before human review. It must not silently tune the signal-region background model to improve agreement.

Phase 4c must then compare the full observed result to both the Phase 4a expectation and the Phase 4b 10% validation result. If data-level nuisance pulls or GoF tests reveal a physics issue traceable to earlier phases, the regression protocol must be triggered. Phase 5 should add final typesetting, figure compositing, more complete comparison plots, and any additional citable calibration or detector variation inputs that can be implemented without violating the reduced-sample scope.

11 Known Limitations and Open Questions

The dominant limitation is missing detector and efficiency correction machinery. The reduced files do not include the full trigger, tau ID, tau trigger, tau energy scale, muon efficiency, pileup, JES/JER, MET, or b-tag variation workflow used in the CMS reference analyses. Phase 4a represents this with a 15% tau/open-data acceptance nuisance and explicit downscope rows, but this is not a substitute for source-specific detector systematics.

The second limitation is missing background and signal components. The expected workspace omits QCD multijet, diboson, single top, W4 or inclusive W+jets, embedded Z to tau tau, associated Higgs, H to WW, and additional DY categories. The largest practical consequence is that the expected background composition is incomplete relative to the CMS papers, especially for fake-tau and electroweak backgrounds.

The third limitation is expected-only blinding. Phase 4a uses background-only Asimov pseudo-data, so the GoF statistic is identically zero by construction and cannot validate real data agreement. This is correct for Phase 4a, but the limitation must be removed by 10% validation and full observed fits in later phases.

The fourth limitation is method sensitivity. Visible mass is robust in reduced files, but CMS JHEP 05 (2014) 104 used stronger mass reconstruction and a broader channel/category program (CMS Collaboration 2014). The current expected limit and discovery diagnostic show that this reduced single-channel model is not expected to observe H to tau tau by itself.

12 Appendix A: Limitation Index

Table 16: Limitation and decision index propagated to Phase 4a. Labels originate in Phase 1 and are tracked through `COMMITMENTS.md`.

Label	Description	Phase 4a status	Impact
[D1]	Search/template-fit strategy	fulfilled	pyhf visible-mass model built
[D2]	Visible mass primary observable	fulfilled	expected fit uses m_{vis}
[D3]	VBF, boosted, zero-jet categories	fulfilled	simultaneous three-category fit
[D4]	W high-mT data normalization	downscoped in 4a	expected-only phase defers real-data CR use
[D5]	Tight tau anti-muon veto	fulfilled	inherited from Phase 3 selection
[D6]	DY/Z 10-15% uncertainty	fulfilled	15% DY norm nuisance
[D7]	pyhf workspace with MC stat terms	fulfilled	25-parameter pyhf model
[D8]	Blinding	fulfilled	real signal-region data not used

Label	Description	Phase 4a status	Impact
[D9]	Alternative observable gate	downscoped	NN and genMET alternatives rejected; add-MET diagnostic only
[L1]	Single-channel scope	carried	sensitivity weaker than CMS all-channel
[L2]	Missing trigger/tau scale factors	carried	15% tau/open-data acceptance nuisance
[L3]	Visible mass weaker than SVFit	carried	expected discovery diagnostic small
[L4]	QCD/multijet unavailable	carried	fake background deferred to sideband validation

13 Appendix B: Numerical Source Map

The AN numerical values are rendered from JSON outputs and external cited records. Result values are not transcribed from prose artifacts. Table 17 maps the main numerical classes to their machine-readable sources.

Table 17: Numerical source map. This table is intended as the audit bridge between the note and the machine-readable artifacts.

Numerical class	Source file	Example fields
Expected CLs limit and discovery diagnostic	phase4_inference/4a_expected/outputs/expected_results.json	expected band, Asimov limit, p-value, Z
Nominal yields and binning	phase4_inference/4a_expected/outputs/nominal_yields.json	bin edges, category yields, S/sqrt(B)
Systematics	phase4_inference/4a_expected/outputs/systematics.json	nuisance sizes, completeness table, downscopes
Signal injection	phase4_inference/4a_expected/outputs/signal_injection.json	injected mu, fitted mu, biases
GoF toys	phase4_inference/4a_expected/outputs/gof_validation.json	toy count, medians, p95, Asimov statistic
Missing components	phase4_inference/4a_expected/outputs/limitations_downscope.json	missing sample list and statuses
Selection raw yields	phase3_selection/outputs/category_yields.json, region_yields.json	category and region counts
Normalization inputs	phase3_selection/outputs/normalization_inputs.json	luminosity, N_gen, cross sections, weights

14 Appendix C: Auxiliary Phase 2 Figures

The Phase 2 figures in this appendix document early feasibility and branch-range checks. They are superseded by Phase 3 final-candidate plots for the selected-event model, but they remain useful provenance for why visible mass, add-MET, transverse mass, jet multiplicity, and VBF variables were considered.

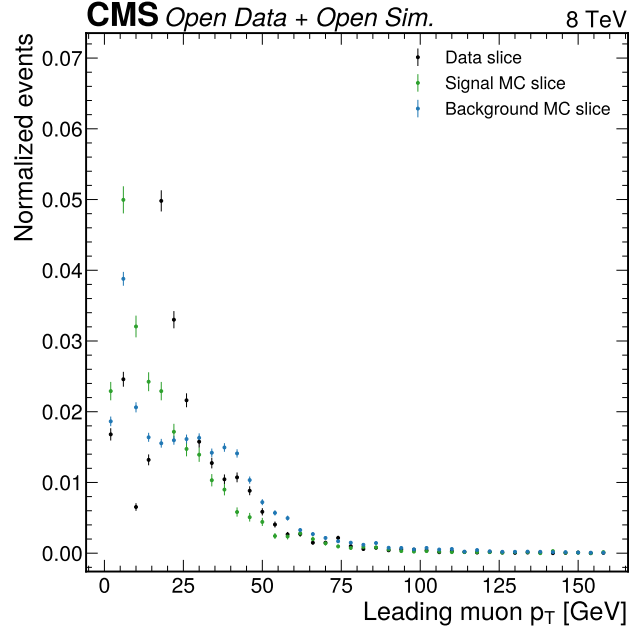


Figure 21: Phase 2 leading muon transverse momentum. This first-candidate slice check validated the presence and range of the muon branches before final object arbitration. It is included as provenance, while the final selected muon distribution in Figure 6 is the analysis-level diagnostic.

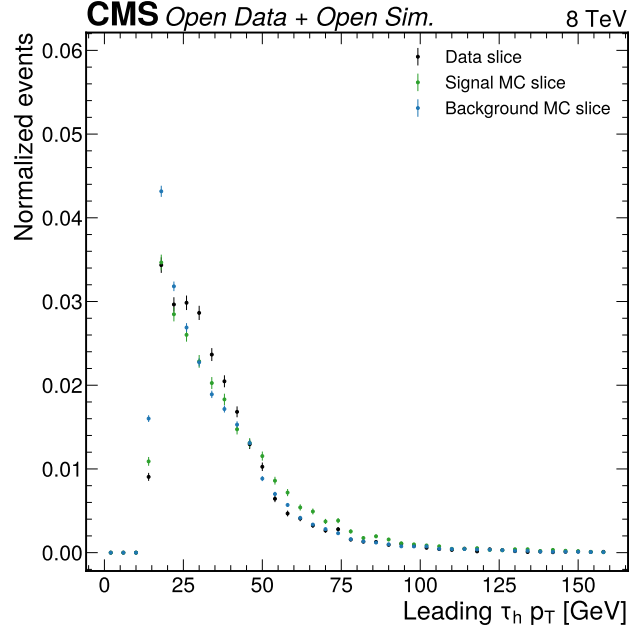


Figure 22: Phase 2 leading tau transverse momentum. This first-candidate plot confirmed that the tau candidate branches are populated in data and MC. The final τ_h selection used for templates is documented by Figure 7.

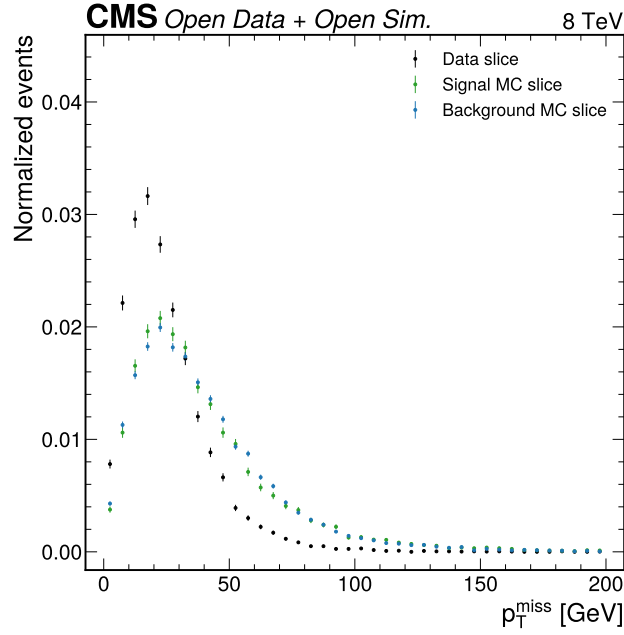


Figure 23: Phase 2 MET diagnostic. This plot established that reconstructed MET is available in the reduced files and can support transverse-mass and add-MET studies. It also helped identify the absence of direct GenMET needed for the requested NN genMET regression.

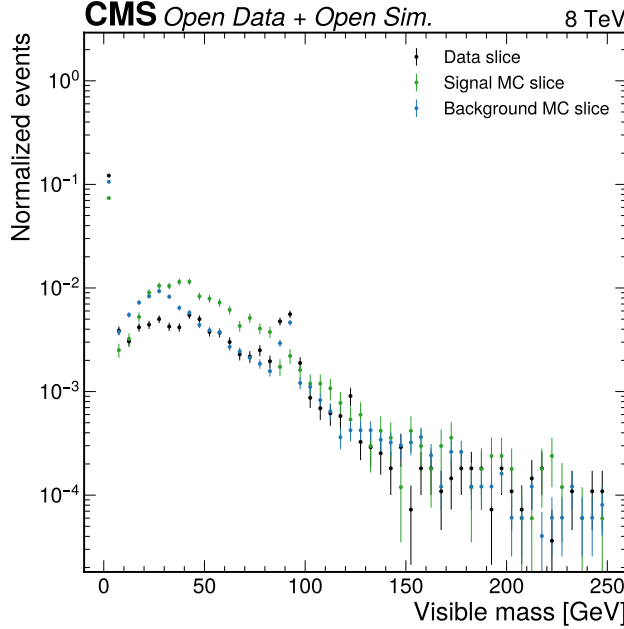


Figure 24: Phase 2 visible-mass diagnostic. This first-candidate visible-mass plot showed that m_{vis} can be constructed but also warned about loose candidate-cleaning artifacts. Phase 3 rebuilt the mass templates after final object selection before Phase 4a used them.

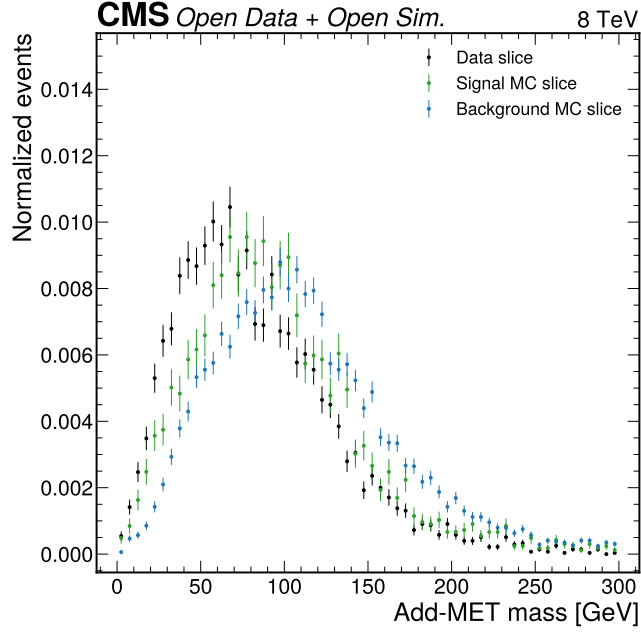


Figure 25: Phase 2 add-MET mass diagnostic. This plot demonstrated that a reconstructed-MET-assisted mass-like observable is available in both data and MC. The observable remains a documented alternative, but it is not the Phase 4a primary fit observable.

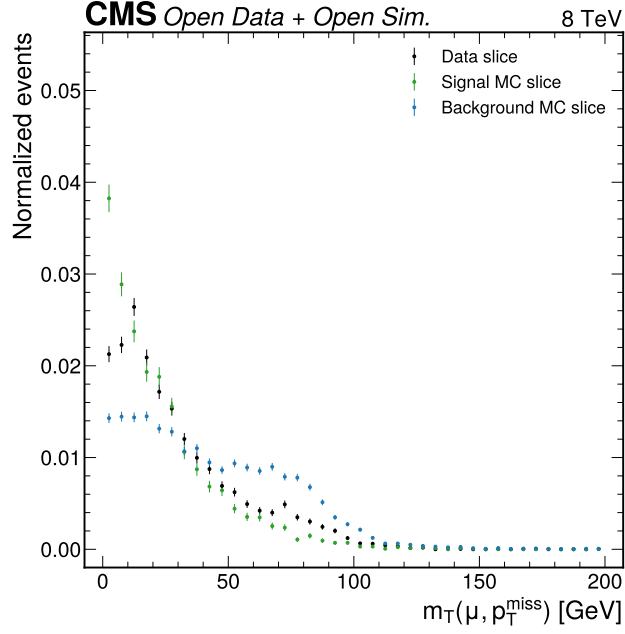


Figure 26: Phase 2 transverse-mass diagnostic. This first-candidate distribution motivated the low- m_T signal-region and high- m_T W control-region definitions. The final selected transverse-mass behavior is shown in Figure 9.

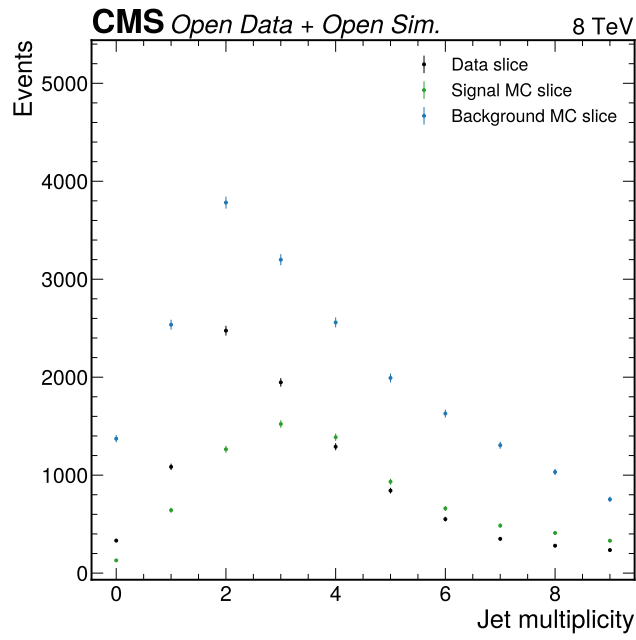


Figure 27: Phase 2 jet multiplicity diagnostic. This plot supported the feasibility of category construction with zero-jet, boosted, and VBF-like topologies. Final clean-jet multiplicity is documented by Figure 11.

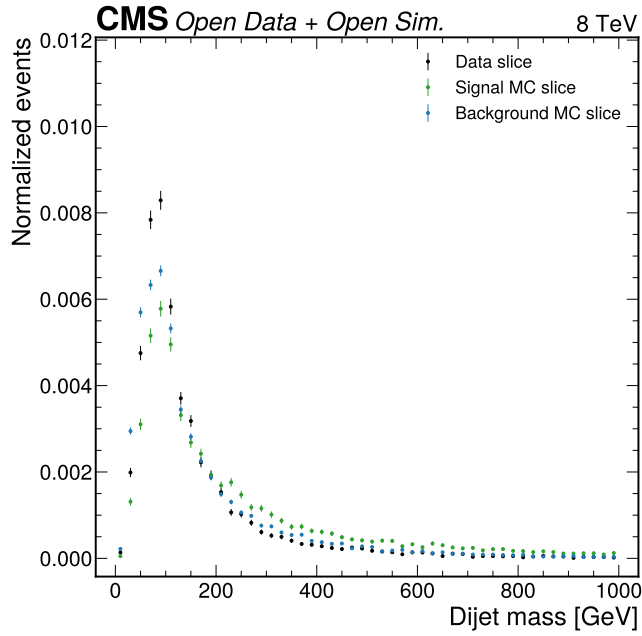


Figure 28: Phase 2 dijet mass diagnostic. The first-candidate survey showed that leading-dijet mass is available and populated enough to study a VBF-like category. The final VBF diagnostic is shown in Figure 12.

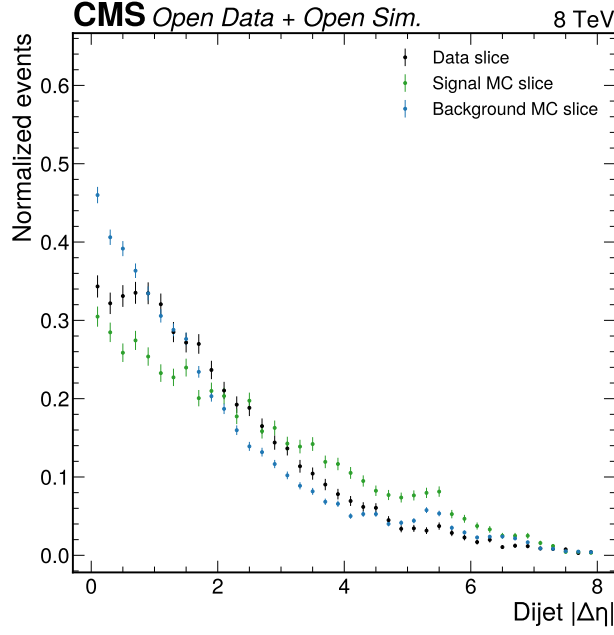


Figure 29: Phase 2 dijet eta-separation diagnostic. This plot identified leading-dijet eta separation as one of the strongest simple discriminators in the early survey. The final selected distribution in Figure 13 supports the retained VBF category.

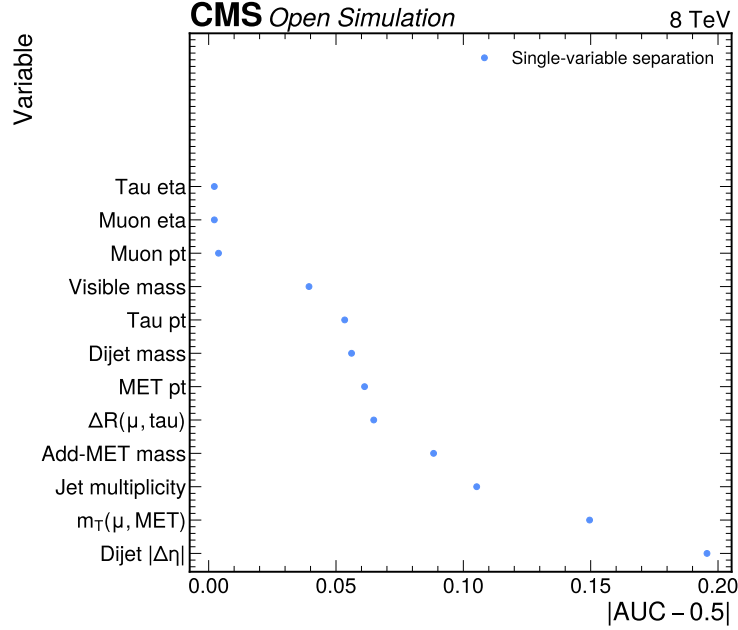


Figure 30: Phase 2 variable separation ranking. The ranking summarized single-variable raw MC separation in the exploratory slice. It explains why VBF-related quantities, transverse mass, jet multiplicity, add-MET mass, and visible mass remained under consideration before Phase 3 validation.

15 Appendix D: Auxiliary Phase 3 Template Figures

The Phase 3 template figures in this appendix are shape-normalized diagnostics produced before the Phase 4a expected model added official Open Data normalization and nuisance parameters. They document the primary visible-mass observable and the add-MET alternative by category.

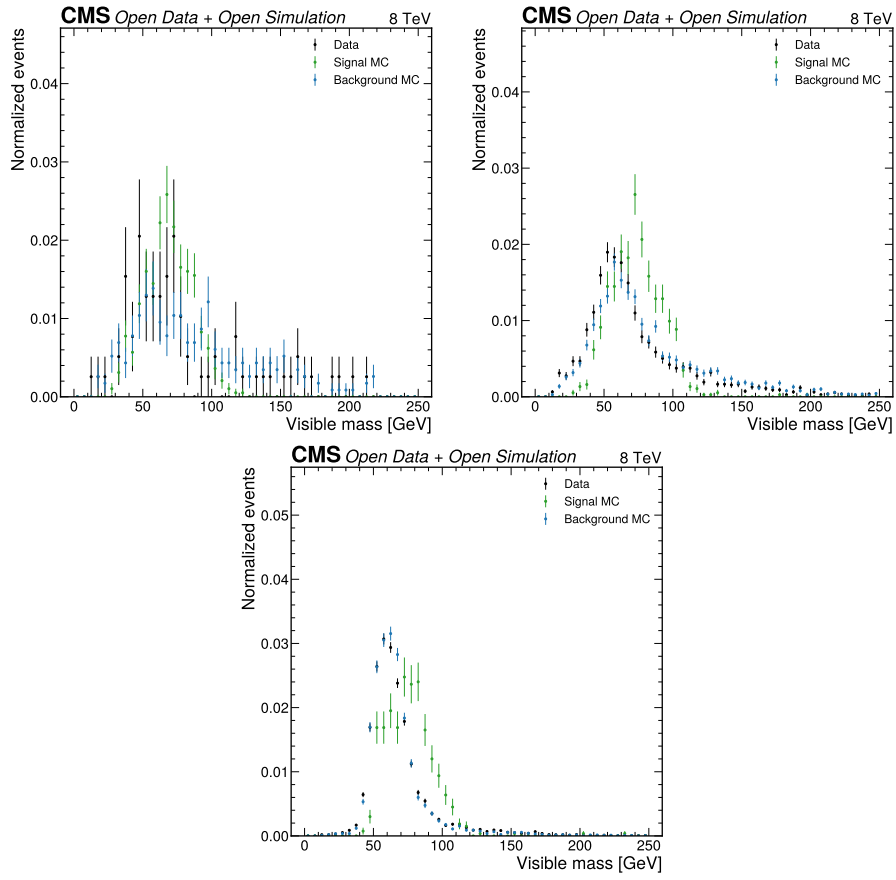


Figure 31: Phase 3 visible-mass diagnostics by category. From left to right, the panels show the VBF, boosted, and zero-jet categories before the Phase 4a expected model added official Open Data normalization. The Phase 4a expected templates in Figures 19, 19, and 19 supersede these normalized diagnostics for the expected statistical model.

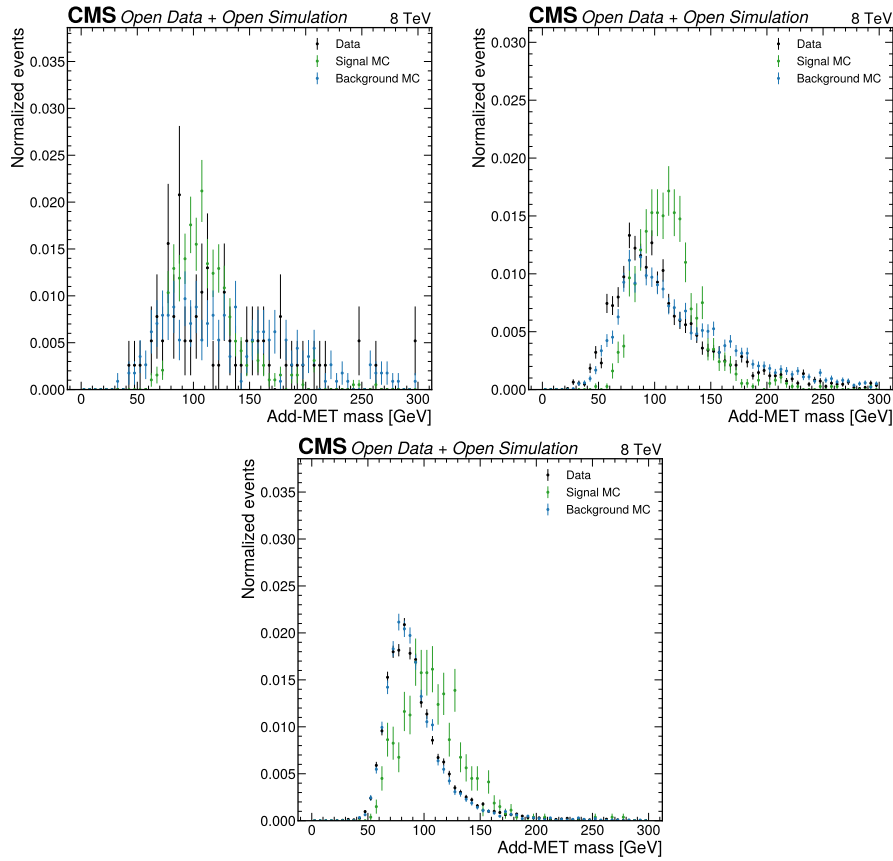


Figure 32: Phase 3 add-MET mass diagnostics by category. From left to right, the panels show the VBF, boosted, and zero-jet categories for the alternative observable retained for audit because add-MET mass was required by the user prompt. The observable supports the documented alternative-observable study but is not promoted over visible mass in Phase 4a.

16 Appendix E: Reproduction Contract

The analysis is managed through pixi. A reproducer should install the environment and then run the phase tasks in order. The Phase 4a expected model is reproducible without looking at observed full-data signal-region results because it uses selected MC templates and background-only Asimov pseudo-data.

Table 18: Reproduction command sequence. Commands and expected outputs are based on `pixi.toml` and the phase artifacts.

Step	Command	Expected outputs
Environment	<code>pixi install</code>	default pixi environment
Phase 2 localization manifest	<code>pixi run phase2-local-manifest</code>	local sample manifest
Phase 2 exploration	<code>pixi run phase2-explore</code> and <code>pixi run phase2-plots</code>	exploration JSON and figures
Phase 3 selection	<code>pixi run phase3-all</code>	selected events, category yields, region yields, figures
Phase 4a model	<code>pixi run phase4a-model</code>	templates, pyhf workspace, expected JSON
Phase 4a plots	<code>pixi run phase4a-plots</code>	expected template and validation figures
Phase 4a validation	<code>pixi run phase4a-validate</code>	JSON, workspace, and figure checks
Full Phase 4a executor chain	<code>pixi run phase4a-all</code>	complete Phase 4a executor outputs

The workflow DAG is: localized ROOT files and Open Data normalization records feed Phase 2 inventory and Phase 3 selection; Phase 3 selected-event summaries feed Phase 4a weighted templates; weighted templates feed the pyhf

workspace; the workspace feeds expected CLs, discovery diagnostics, injection tests, GoF toys, and Phase 4a figures; this note renders the JSON and figure artifacts into analysis prose.

17 Appendix F: Note Writer Self-Check

This self-check is updated by the note writer before handoff to the typesetter. The source files are `ANALYSIS_NOTE_4a_v1.md`, `references.bib`, and the JSON files listed in Table 17.

Table 19: Self-check targets and current status. The final counts are reported in the note-writer final status message after mechanical verification.

Check	Target or rule	Status
Display equations	at least 8	drafted above target
Figure references	at least 20	drafted above target
Table references	at least 5	drafted above target
Implemented systematic subsections	4 of 4	complete
Expected-only blinding statement	required	present in introduction, results, and limitations
Figure file existence	all referenced PDFs	verified after drafting
Numerical source consistency	key JSON values match rounded note values	verified after drafting

References

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